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The Editorial Board
Topology and its Applications
Elsevier

Dear Editors,

I am submitting the manuscript “**Homology Class Asymmetry in the Loxodromic Twist Spectrum of Compact Hyperbolic 3-Manifolds**” for consideration in *Topology and its Applications*.

Summary of results. We introduce a homology-resolved extreme-value invariant of the complex length spectrum of a compact hyperbolic 3-manifold: the *spectral floor* $\phi_{\text{floor}}^{(k)}(L)$, defined as the minimum folded loxodromic twist angle in homology class $k \in H_1(M; \mathbb{Z}/5)$ among words of length $\leq L$. For the manifold *m006* (OrientableClosedCensus[43], $H_1 = \mathbb{Z}/5$, $\text{vol} = 2.029$), a complete computational census of 1,062,802 loxodromic words up to length 12 reveals that the five floors decrease exponentially with L but at class-dependent rates forming a paired hierarchy: $c_1 = c_4 \approx 0.985$, $c_2 = c_3 \approx 0.401$, $c_0 \approx 0.750$. AIC model selection confirms the exponential law over power-law alternatives ($\Delta\text{AIC} > 40$ for three of five classes). The same pairing structure appears in the Meyerhoff manifold *m003* (also $H_1 = \mathbb{Z}/5$), suggesting the hierarchy is a property of $\mathbb{Z}/5$ torsion rather than of either specific manifold. We conjecture that the pairing $c_k = c_{5-k}$ reflects complex conjugation in the invariant trace field.

Why *Topology and its Applications*. The paper introduces a new invariant of the complex length spectrum, proves its basic properties, and provides extensive computational evidence for a structural conjecture. It sits at the intersection of hyperbolic 3-manifold topology, computational geometry (SnapPy), and arithmetic group theory — exactly the scope of TAIA. The methods are elementary (word enumeration, homology class computation, AIC model selection) and the results are fully reproducible from publicly available software.

Prior submissions. The manuscript was previously submitted to *Annales de l’Institut Fourier* (rejected for editorial fit, April 2026) and *Geometriae Dedicata* (rejected for capacity, April 2026; the editors noted the paper was “mathematically sound and interesting”). It has not been submitted elsewhere and is not currently under review at any other journal. A preprint is available at SSRN (Abstract ID: 6688021).

Suggested reviewers. Given the computational and arithmetic nature of the results, I suggest the following potential reviewers (I have no conflict of interest with any of them):

Nathan Dunfield (UIUC), Alan Reid (Rice), Marc Lackenby (Oxford).

Sincerely,

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